

Cu6 ϕ_1 $\rightarrow$

$$\begin{aligned}
& \frac{1}{16\pi^2} \left(\frac{1}{72} g_3 g_1^2 y_u^{i1i2} LF_{2,1,0} [m_1, m_q^{i1}] - \frac{1}{36} g_3 g_1^2 y_u^{i1i2} LF_{3,1,-1} [m_1, m_q^{i1}] + \frac{1}{72} g_3 g_1^2 y_u^{i1i2} \right. \\
& LF_{4,1,-2} [m_1, m_q^{i1}] + \frac{2}{9} g_3 g_1^2 y_u^{i1i2} LF_{2,1,0} [m_1, m_u^{i2}] - \frac{4}{9} g_3 g_1^2 y_u^{i1i2} LF_{3,1,-1} [m_1, m_u^{i2}] + \\
& \frac{2}{9} g_3 g_1^2 y_u^{i1i2} LF_{4,1,-2} [m_1, m_u^{i2}] + \frac{3}{8} g_3 g_2^2 y_u^{i1i2} LF_{2,1,0} [m_2, m_q^{i1}] - \\
& \frac{3}{4} g_3 g_2^2 y_u^{i1i2} LF_{3,1,-1} [m_2, m_q^{i1}] + \frac{3}{8} g_3 g_2^2 y_u^{i1i2} LF_{4,1,-2} [m_2, m_q^{i1}] - \\
& \frac{1}{12} g_3^3 y_u^{i1i2} LF_{2,1,0} [m_3, m_q^{i1}] - \frac{7}{12} g_3^3 y_u^{i1i2} LF_{3,1,-1} [m_3, m_q^{i1}] + \\
& \frac{2}{3} g_3^3 y_u^{i1i2} LF_{4,1,-2} [m_3, m_q^{i1}] - \frac{1}{12} g_3^3 y_u^{i1i2} LF_{2,1,0} [m_3, m_u^{i2}] - \\
& \frac{7}{12} g_3^3 y_u^{i1i2} LF_{3,1,-1} [m_3, m_u^{i2}] + \frac{2}{3} g_3^3 y_u^{i1i2} LF_{4,1,-2} [m_3, m_u^{i2}] + \\
& \frac{1}{4} g_3 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{2,1,0} [\tilde{\mu}, m_d^r] - \frac{1}{2} g_3 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{3,1,-1} [\tilde{\mu}, m_d^r] + \\
& \frac{1}{4} g_3 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{4,1,-2} [\tilde{\mu}, m_d^r] + \frac{1}{2} g_3 \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{2,1,0} [\tilde{\mu}, m_q^p] - \\
& g_3 \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{3,1,-1} [\tilde{\mu}, m_q^p] + \frac{1}{2} g_3 \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{4,1,-2} [\tilde{\mu}, m_q^p] + \\
& \frac{1}{4} g_3 \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{2,1,0} [\tilde{\mu}, m_u^r] - \frac{1}{2} g_3 \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{3,1,-1} [\tilde{\mu}, m_u^r] + \\
& \frac{1}{4} g_3 \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{4,1,-2} [\tilde{\mu}, m_u^r] - \frac{1}{9} g_3 m_1 g_1^2 a_u^{i1i2} LF_{2,1,1,0} [m_1, m_q^{i1}, m_u^{i2}] + \\
& \frac{1}{9} g_3 m_1 g_1^2 a_u^{i1i2} LF_{3,1,1,-1} [m_1, m_q^{i1}, m_u^{i2}] - \frac{1}{12} g_3 g_1^2 y_u^{i1i2} LF_{1,1,1,0} [m_1, m_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{6} g_3 g_1^2 y_u^{i1i2} LF_{2,1,1,-1} [m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{12} g_3 g_1^2 y_u^{i1i2} LF_{3,1,1,-2} [m_1, m_q^{i1}, \tilde{\mu}] + \\
& \frac{1}{3} g_3 g_1^2 y_u^{i1i2} LF_{1,1,1,0} [m_1, m_u^{i2}, \tilde{\mu}] - \frac{2}{3} g_3 g_1^2 y_u^{i1i2} LF_{2,1,1,-1} [m_1, m_u^{i2}, \tilde{\mu}] + \\
& \frac{1}{3} g_3 g_1^2 y_u^{i1i2} LF_{3,1,1,-2} [m_1, m_u^{i2}, \tilde{\mu}] - \frac{1}{12} g_3 g_1^2 y_u^{i1i2} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{1}{3} g_3 g_1^2 y_u^{i1i2} LF_{2,2,1,-2} [m_1, \tilde{\mu}, m_u^{i2}] + \frac{3}{4} g_3 g_2^2 y_u^{i1i2} LF_{1,1,1,0} [m_2, m_q^{i1}, \tilde{\mu}] - \\
& \frac{3}{2} g_3 g_2^2 y_u^{i1i2} LF_{2,1,1,-1} [m_2, m_q^{i1}, \tilde{\mu}] + \frac{3}{4} g_3 g_2^2 y_u^{i1i2} LF_{3,1,1,-2} [m_2, m_q^{i1}, \tilde{\mu}] + \\
& \frac{3}{4} g_3 g_2^2 y_u^{i1i2} LF_{2,2,1,-2} [m_2, \tilde{\mu}, m_q^{i1}] + \frac{1}{6} m_3 g_3^3 a_u^{i1i2} LF_{2,1,1,0} [m_3, m_q^{i1}, m_u^{i2}] + \\
& \frac{4}{3} m_3 g_3^3 a_u^{i1i2} LF_{3,1,1,-1} [m_3, m_q^{i1}, m_u^{i2}] + \frac{1}{6} g_3 g_1^2 y_u^{i1i2} LF_{2,1,1,-1} [\tilde{\mu}, m_1, m_q^{i1}] - \\
& \frac{1}{12} g_3 g_1^2 y_u^{i1i2} LF_{3,1,1,-2} [\tilde{\mu}, m_1, m_q^{i1}] - \frac{2}{3} g_3 g_1^2 y_u^{i1i2} LF_{2,1,1,-1} [\tilde{\mu}, m_1, m_u^{i2}] + \\
& \frac{1}{3} g_3 g_1^2 y_u^{i1i2} LF_{3,1,1,-2} [\tilde{\mu}, m_1, m_u^{i2}] - \frac{3}{2} g_3 g_2^2 y_u^{i1i2} LF_{2,1,1,-1} [\tilde{\mu}, m_2, m_q^{i1}] + \\
& \frac{3}{4} g_3 g_2^2 y_u^{i1i2} LF_{3,1,1,-2} [\tilde{\mu}, m_2, m_q^{i1}] - \frac{1}{2} g_3 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{2,1,1,0} [\tilde{\mu}, m_d^r, m_q^p] + \\
& \left. \frac{1}{2} g_3 \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{3,1,1,-1} [\tilde{\mu}, m_d^r, m_q^p] \right)
\end{aligned}$$